Bundling an Executable

One-Folder Approach

Activate the `gui` virtual environment.

```
conda activate gui
```

Now that we have a GUI application, it is time to transfrom it into an executable with Pyinstaller. Start by installing Pyinstaller with Conda. Remember, you need to fetch the package from the `conda-forge` channel.

```
conda install pyinstaller -c conda-forge -y
```

We can use Pyinstaller to bundle a Kivy application as an executable. There is a slight change when compared to what we did in the previous assignment. Since our entire Kivy application is contained in a single Python file we do not need to tell Pyinstaller where to look for additional files. We could run `pyinstaller app.py` and create our executable.

However, this means our executable will be named `app` which is not very descriptive. Instead, create a spec file and change the name to `counter`. Since this is a windowed application, use the `-w` flag so your computer will not open a terminal when the executable runs (this assumes you are double-clicking on an icon).

```
pyi-makespec --name counter app.py -w
```

Run Pyinstaller with the newly created spec file to generate the executable.

```
pyinstaller counter.spec
```

By default, Pyinstaller uses the one-folder approach. So the executable will be in the `counter` directory which is in the `dist` directory. Use `./` followed by the path to the executable. Click the link to see the output. Your GUI application should work just as before.

```
./dist/counter/counter
```

Preview the GUI

One-File Approach

We can also use the one-file approach when bundling an executable. First, let's delete the `build` and `dist` directories as well as the spec file.

```
rm dist build -r counter.spec
```

Generate the spec file just as before, but add the `--onefile` flag.

```
pyi-makespec --name counter app.py -w --onefile
```

Just as before, generate the executable by running Pyinstaller with the newly created spec file.

```
pyinstaller counter.spec
```

The single executable file `counter` should reside in the `dist` directory. Run it with `./` and the path to the executable. Click the link to see the output.

```
./dist/counter
```

Preview the GUI

Lastly, deactivate the `gui` virtual environment.

```
conda deactivate
```

Reading Question

How do you differentiate a windowed application from a command line application in Pyinstaller?

- ( ) You do not need to do anything; Pyinstaller can figure this out automatically.
- (•) Use the `-w` flag when bundling or creating the spec file.
- ( ) Use the `--gui` flag when bundling or creating the spec file.
- ( ) Use the `--kivy` flag when bundling or creating the spec file.

When bundling a GUI application with Pyinstaller, we do not want the application to open the terminal on launch. Use the `-w` flag to tell Pyinstaller this is a windowed application.

Check It

Next